Timestamp	Username	mAP	Approach
11/11/2025 1:36:16	Dario Liotta - Gabriel Amorosetti	0.911311	We first implemented a custom resizing function that preserved the original aspect ratio of the images. Using the FLANN algorithm, we initially performed a grid search to find the optimal maximum size for resizing images in the dataset. Next, we conducted another grid search over the threshold ratio to select the best value. Afterward, we applied random searches to identify the most promising intervals for descriptor parameters (contrastThreshold, edgeThreshold, sigma, nOctaveLayers) and FLANN parameters (number of trees and number of checks). Finally, using the refined intervals, we performed a last random search to determine the overall best configuration.
11/11/2025 17:15:28	Lazzari Tommaso	0.901	I built one SIFT descriptor for each query image and I fine tuned all the hyperparameters required by the cv2.SIFT_create() function; moreover I fine tuned also the threshold for the ratio test
11/12/2025 22:38:34	zahraasaleh	0.897	I changed the code to use different SIFT parameters for each brand, normalized the descriptors, used KNN matching with ratio test, distance limit, symmetry check, spatial spread, and RANSAC, and increased the number of recall points when computing APs. Then I calculated a score for each query–database pair instead of just counting matches.
11/10/2025 11:27:55	Hooman Sabzi	0.894	Optimization of SIFT hyperparameters via random search, refining search intervals around the best-performing values, followed by feature matching using a FLANN-based matcher for efficient nearest-neighbor search.
11/12/2025 17:06:06	Giovanni24	0.892	I only focused on improving Coca Cola, with the same code. I just changed the parameters nOctaveLayers, contrastThreshold, and edgeThreshold, sometimes randomly, for a very long time
11/12/2025 22:15:38	Alessandro Murru	0.886	I performed a Random Search to optimize the parameters contrastThreshold, nOctaveLayers, and edgeThreshold, which are global to the detector, as well as the ratio_thresh parameter, for which a distinct value was optimized for each individual logo.
11/12/2025 22:19:08	Alessandro Murru	0.886	I performed a Random Search to optimize the parameters contrastThreshold, nOctaveLayers, and edgeThreshold, which are global to the detector, as well as the ratio_thresh parameter, for which a distinct value was optimized for each individual logo.
11/11/2025 21:51:15	Picello Davide + Fantin Anthony	0.885	Manual parameter configuration was replaced with a two-stage Bayesian Optimization process. First, an optimization was run to find the optimal SIFT parameters. Lately, to further refine the results, these parameters were locked while a second optimization found the best ratio threshold for each individual query logo.
11/11/2025 14:50:36	D'Ovidio Giacomo, Furno Francesco	0.884	SIFT was applied separately on each channel (gs, r, g, b) and the resulting descriptors were concatenated in a 512D vector. Parameters were tuned for improved detection. Matching used Lowe's ratio test and RANSAC-based affine verification to retain only consistent inliers.
11/11/2025 16:53:53	Mahsa Ekrahi	0.883	Performed a random search with FLANN to optimize the parameters of the SIFT algorithm. Starting with the full set of parameters, evaluated performance iteratively by eliminating parameters one at a time and re-assessing results. Additionally, implemented an adaptive threshold for each logo, determined through manual search.
11/11/2025 21:14:38	Pankhuri and Sofia	0.882	We ran a bayesian optimizer to optimize the SIFT parameters, along with mutual matching and RANSAC for geometric verification. The evaluation is done on the basis of precision-recall and AP values as mentioned.
11/12/2025 21:03:10	Alfeo	0.8815	My best approach has maintained a SIFT detector combined with a brute-force k-nearest neighbor matcher (k=2). A grid search followed by a refined search around the optimum was performed over the SIFT parameters (nOctaveLayers, contrastThreshold, edgeThreshold). The Lowe's ratio threshold was increased to 0.85.

Timestamp	Username	mAP	Approach
11/11/2025 16:22:02	Elio Shyti, Max Horn	0.879	We tuned key SIFT parameters (nOctaveLayers, contrastThreshold, edgeThreshold, sigma, ratio_thresh) using an arbitrary set of parameters for grid search and selected the setup with the highest mean average precision (mAP = 0.879).
11/12/2025 15:04:32	zahraaslh	0.872	Added different SIFT parameters for each logo, used RootSIFT normalization for more stable descriptors, applied ratio and distance filters with symmetry check, included RANSAC geometric verification, KNN matching and replaced simple match counting with a weighted scoring based on inliers and match quality.
11/12/2025 22:41:00	Frances Ferrabee, Emilija Ivkovic	0.871	we tuned the hyperparameters in SIFT for each logo, We added RANSAC , used different norms L1 and L2
11/10/2025 19:04:25	Troiani	0.87	Implemented FL-ANN based matcher and performed bayesian optimization over all relevant pipeline search space (SIFT hyperparameters, FLANN parameters, Lowe Ratio...). Tried to implement weigthed reciprocal rank fusion (WRRF), a common practice in information retrieval to reduce variance like in ensemble methods, but yielded worse results.
11/11/2025 21:52:41	Sara Pangrazio	0.865	I optimized the SIFT-based pipeline using grid search, then applied Brute-Force mutual matching followed by geometric verification through homography (RANSAC).
11/11/2025 20:32:36	Davide Sut	0.864	SIFT and AKAZE combined, dynamic ratio test and geometric transformation
11/12/2025 10:07:27	Marco Diotto	0.861	I have used a similar approach to my other submission (I have used rootSIFT in order to increase robustness. I have utilized different thresholds for the Lowe's ratio test for each logo, these thresholds were chosed with the objective of improving the precision/recall tradeoff. Finally I have applied different versions of RANSAC with different thresholds for each logo in order to find the best geometric description for each category.) but with some changes. I wanted to increase the speed matching using FLANN, however, this ended up giving better results than BF matcher (with the right choice of FLANN/SIFT parameters). Then i had to modify the RANSAC parametrization to fit this new pipeline. One major difference is that for the Marlboro class i added a fraction of the matches to the sum of inliers, in order to "boost" the probability of assigning the Marbloro class, which was the most overlooked one
11/11/2025 15:48:28	Giulio Morandini (2139632) Marco Ve	0.858	we decided to tune the SIFT parameters through a grid search, also we extract some features using greyscale images and add a RANSAC-based geometric verification step that counts only inlier matches after the ratio test.
11/12/2025 22:28:06	Arturo Bevilacqua	0.858	"Fine-tuned" different combinations of hyperparameters (i.e. SIFT parameters + BFmatcher norm + Ratio Threshold) for each logo separately.
11/10/2025 9:53:14	Matteo Garato	0.852	I did a hyperparameters grid search using SIFT as descriptor and FLANN as matcher. The optimal configuration was found with the following hyperparameters: {'nfeatures': 500, 'contrastThreshold': 0.06, 'edgeThreshold': 5, 'sigma': 1.2, 'ratio_thresh': 0.7}. I measured the execution time of the grid search in order to compare it with other techniques, it was 1734 seconds. The results obtained by using SIFT as descriptor and the Brute Force as matcher were mAP = 0.830 and time = 1938, it means that FLANN leads to better computational performance (since it uses approximated results) and the result in term of performance are similar. For comparison, I also tested ORB instead of SIFT: this significantly improved the execution time, but the performance was worse (mAP = 0.573).
11/10/2025 17:15:36	Fusari, Micino	0.851	Img_size = (480, 480). Ratio_thresh = 0.72. Detector = cv2.SIFT_create(nfeatures=600).
11/12/2025 9:49:00	Marco Diotto	0.849	I have used rootSIFT in order to increase robustness. I have utilized different thresholds for the Lowe's ratio test for each logo, these thresholds were chosed with the objective of improving the precision/recall tradeoff. Finally I have applied different versions of RANSAC with different thresholds for each logo in order to find the best geometric description for each category.

Timestamp	Username	mAP	Approach
11/12/2025 16:49:09	Mattia Pirdeni	0.847	Optimize SIFT parameters via a grid search, and integrate RANSAC with cv2.findHomography to reject outlier matches. Best parameters: Best Parameters: nfeatures: 1000 nOctaveLayers: 3 contrastThreshold: 0.03 edgeThreshold: 10 sigma: 1.6 ratio_thresh: 0.7 ransac_reproj_threshold: 3.0 ransac_confidence: 0.99
11/12/2025 16:44:08	Nicole Mietto, Teresa Marangon	0.846	our approach: -FEATURE DETECTION: we used a sift with different parameters for each logo to enhance each logo characteristics individually. For this reason we created a nested list of database descriptor and keypoints in order to compare them with the characteristic descriptor of the logo. Custom SIFT Parameters: *cv2.SIFT_create() for coca cola *cv2.SIFT_create(edgeThreshold=20) for Heineken *cv2.SIFT_create(edgeThreshold=15) for Marlboro *cv2.SIFT_create(nOctaveLayers=4,contrastThreshold=0.03, sigma = 1.2) for Starbucks. -FEATURE MATCHING: we used cv2.BFMatcher_create(cv2.NORM_L2) in order to capture all possible correlations between matches. We selected matches with knnMatch with k=2 and keep the ones with a ratio <= threshold (logo-specific thresholds: [0.78, 0.74, 0.765, 0.713]). -OUTLIER ELIMINATION: Then we compute the homography filter, with RANSAC method, to eliminate the outliers matches. Algorithm was applied only if the number of initial filtered matches exceeded a fixed, logo-specific threshold ([14,4,10,4]).
11/12/2025 11:02:39	Mattia Ponchio	0.844	I tried a different threshold for each logo and then selected the best from each one. Moreover, I tried the FLANN-based matcher, but then discarded it due to its stochastic nature since the speedup was not relevant for the provided data size.
11/11/2025 15:02:40	2141895 (Diana Andrade) - 2141896 (	0.841	We first did some hyperparameter tuning of the SIFT detector, and the ratio test's threshold. We compared our results to the brute force approach using only the euclidean distance, but we noticed the results were better using the ratio test. Finally, we applied some geometrical transformation to visualize how good our matchings were.
11/12/2025 20:16:37	Tharen Candi	0.835	bayesian search over params for sift and matching with ransac geo verification
11/10/2025 9:48:28	Matteo Garato	0.83	I did a hyperparameters grid search using SIFT as descriptor and the Brute Force as matcher. The optimal configuration was found with the following hyperparameters: {'nfeatures': 500, 'contrastThreshold': 0.04, 'edgeThreshold': 5, 'sigma': 1.2, 'ratio_thresh': 0.7}. I measured the execution time of the grid search in order to compare it with other techniques, it was 1938 seconds. For comparison, I also tested ORB instead of SIFT: this significantly improved the execution time, but the performance was worse (mAP = 0.573).
11/11/2025 13:02:55	marinalimabraga	0.829	I created a spreadsheet with a table containing all the parameters I could change to try better results. I did around 40 trials and this was my best one so far. I saw that the best macther was the FLANN so I replace the BF. Also, I tried using the methods of inliers instead of matches in the final ranking and with that I got better results, utilizing the RANSAC one. Also, in the parameters I kept changing checks (FLANN),trees (FLANN),Ratio,nOctaveLayers,ContrastThreshold,EdgeThreshold,nfeature, ransacThr until I got the best match between all of them.
11/11/2025 22:38:59	ArturoBevilacqua	0.827	Tuning of the ratio threshold for each logo individually (with final ratio_thresh = [0.78, 0.71, 0.67, 0.83])
11/11/2025 1:46:57	Evgenii Markin	0.824	Increased max number of features, slightly changed other SIFT parameters. Added symmetry check q -> db and db -> q
11/12/2025 19:22:58	Andrea Tonellato	0.823	I did a grid search over the parameters (nfeatures, contrastThreshold, edgeThreshold) of the cv2.SIFT_create() function for SIFT implementation and the ratio_thresh value in order to maximize the mAP.
11/11/2025 20:19:18	Alice Mameli	0.822	Hyperparameters selected with grid search in SIFT model.(nfeatures=400, nOctaveLayers=4, contrastThreshold=0.03, edgeThreshold=8, sigma=1.5)

Timestamp	Username	mAP	Approach
11/11/2025 9:29:54	Giada Baro	0.821	I applied the FLANN-base matcher
11/11/2025 15:57:13	Lisa Arduino	0.814	First, I configured SIFT to increase keypoint coverage while keeping descriptors stable: nfeatures = 10 000 ensures sufficient keypoints on high-resolution or cluttered images so we don't under-sample the scene, nOctaveLayers = 3 ensures balanced scale sampling and keeps runtime reasonable while covering multi-scale structure, contrastThreshold = 0.02, lower than default, admits more low-contrast keypoints, edgeThreshold = 10 Default value and sigma = 1.5, slightly higher initial blur, reduces noise and stabilize descriptors at common image scales. Then, I applied a grid search to fine-tune the Lowe's ratio threshold within the range [0.6, 0.9]. The best results were obtained at ratio = 0.85. A ratio of 0.85 is relatively permissive, meaning it retains a larger number of matches, including some weaker correspondences.
11/11/2025 16:53:21	Marc Gollinucci	0.813	used FLANN based matcher instead of BF, hyperparameter tuning for: 'nOctaveLayers': 3, 'contrastThreshold': 0.05, 'edgeThreshold': 11, 'sigma': 1.6, 'ratio_thresh': 0.75, 'ransac_thresh': 3.0, 'FLANN_INDEX_KDTREE': 1, 'checks': 150
11/11/2025 9:39:13	Marco Augusto Carturan, Andrea Calz	0.812	The main changes we applied to the original notebook are: 1) usage of the GRID SEARCH in order to retrieve the best combination of "ratio_threshold" and "ransac_reprojection"; 2) Normalization of the score dividing the inlier count by the keypoints query count.
11/10/2025 10:11:56	2142932	0.806	Use a grid search for the parameters: nfeatures, contrastThreshold, edgeThreshold. Performed the homography-aligned and calculate the mAP for all possible combination and take the maximum mAP.
11/11/2025 14:35:53	Riccardo Niccolò Agosti	0.806	Tuned the SIFT algorithm's nfeatures, contrastThreshold, and edgeThreshold parameters using a grid search, and also incorporated homography alignment for more robust feature matching.
11/12/2025 0:00:45	Adrian Valentin Josef Seidl	0.801	I optimized the parameters and I also implemented the option to use to Flann optimizing also here the parameters..
11/12/2025 20:58:12	sobhan hosseini	0.8	Starting from SIFT + ratio test baseline, we (1) resized images to 800×800 to increase keypoints on thin logo strokes, (2) applied RootSIFT normalization, (3) used FLANN (KD-tree) with Lowe's ratio 0.70 to prefilter candidates, and (4) performed spatial verification only on the top-K candidates with a tight RANSAC threshold (1.8–2.0 px). To improve robustness to small viewpoint/blur differences (notably for Coca-Cola's script), we evaluated multiple micro-augmentations of the query (±10° rotation, slight blur/scale) and kept the best RANSAC inlier count per database image. Final ranking score = inliers + 1e-3 × (#good). This raised AP on the hardest query while keeping the strong classes stable.
11/12/2025 19:40:08	Elia Carta	0.797	I applied cross search for the hyperparameters of SIFT and FLANN, I tried also different values for the image size and the ratio threshold
11/12/2025 10:12:53	Meritxell Pena	0.796	First I replaced the baseline Brute-Force Matcher with the FLANN-based matcher. To further filter out incorrect matches, I introduced geometric consistency verification using a homography estimation with RANSAC. To improve a little bit the result, I have changed the hybrid normalized score combining inliers (main signal) and matches (tie-breaker), normalized by the number of keypoints in the query image.
11/12/2025 22:14:46	Daniele Gelmini	0.793	I replaced the default SIFT detector with a custom-tuned version (contrastThreshold=0.02, nfeatures=1000) to detect more keypoints. To manage this larger set of features, I then implemented a high-precision FlannBasedMatcher (checks=100). Then I added a RANSAC geometric filter (cv2. findHomography) to discard impossible matches and ensured my final score was based only on geometrically coherent inliers. Finally, I stabilized this result by setting a random seed (cv2.setRNGSeed (1)) to remove the variability from RANSAC, obtaining 0.793 mAP score.

Timestamp	Username	mAP	Approach
11/11/2025 18:27:41	Riccardo Milan	0.785	The first approach I used, is simply to tune the SIFT modifying the default parameters: I experimented with nOctaveLayers, contrastThreshold and edgeThreshold. The best combination I found is nOctaveLayers=3 (the default), lowered contrastThreshold to 0.02 (default 0.04), and lowered edgeThreshold to 8 (default 10). This aims to detect more keypoints, especially on lower-contrast database images.
11/11/2025 18:29:55	Milan Riccardo	0.77	In the third approach, I experimented geometric verification as suggested in the contest slides. After obtaining initial "good matches" from the SIFT + Ratio Test baseline, I applied a more robust filtering step. I used cv2.findHomography with the RANSAC algorithm to compute the perspective transformation between the query and database images. Matches that are inconsistent with this global transformation are rejected. The final score is the count of RANSAC inliers, providing a much more reliable measure of a true match than the simple count of ratio-tested features.
11/12/2025 17:20:16	Iaroslav Tkachenko	0.77	I kept the dataset, GT, and SIFT features unchanged, but replaced the baseline scoring (BF + Lowe's ratio count) with geometric verification. Specifically, I used SIFT + FLANN (KD-Tree, checks=50) with Lowe's ratio = 0.70 to select tentative matches, then estimated a homography with RANSAC (reprojection threshold = 3 px) and used the number of inliers as the ranking score for each DB image. I computed AP per logo and mAP across the four queries. This improved overall mAP from 0.758 to 0.770.
11/11/2025 20:43:52	Beatrice Riello	0.76	parameter tuning
11/9/2025 21:18:09	Baseline	0.759	Default approach and parameters (i.e. standard SIFT detector & descriptors, 2nn-ratio test with standard threshold).
11/11/2025 18:28:51	Milan Riccardo	0.665	In my second approach I replaced SIFT entirely with ORB. I tried various number of features to ensure a sufficient number of keypoints. Crucially, since ORB produces binary descriptors, I used the NORM_HAMMING distance as matching strategy, which is the correct metric for comparing binary strings. The best number of features I found to be is 2900.